Sambit Kumar Barik

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Education

- **National Institute of Technology, Rourkela** — Rourkela, India
  Bachelor of Technology - Mining Engineering — *December 2020 - August 2024*
  Courses: *Machine Learning, Deep Learning, Data Structures, Algorithms, Databases, Operating Systems, Compiler Design*

Skills Summary

- **Languages**: **Python**, **Bash**, **Rust**, **C++**
- **AI/ML Stack**: **PyTorch**, **TensorFlow/Keras**, **ONNX**
- **LLM/RAG/Agentic Frameworks**: **Transformers**, **LangChain**, **LangGraph**, **LlamaIndex**, **DSPy**, **GraphRAG**, **Hybrid Retrieval (RRF)**, **Rerankers**, **Ragas**, **LangSmith**, **Firecrawl**
- **Vector/Graph DB**: **Qdrant**, **Milvus**, **LanceDB**, **Neo4j**, **PostgreSQL/pgvector**
- **Computer Vision**: **OpenCV**, **Pytesseract**, **TensorRT**, **OpenVINO**
- **Infrastructure/Observability**: **Docker**, **Kubernetes/Helm**, **Kafka**, **vLLM**, **Triton**, **Temporal**, **CUDA/cuDNN**, **OpenTelemetry**, **Prometheus/Grafana**, **W&B**
- **Development**: **Git**, **FastAPI**, **VSCode**, **Colab**
- **Deployment**: **Linux**, **AWS**, **CI/CD**

Experience

- **Skylark Labs** — Pune, Maharashtra, India
  Machine Learning Engineer - I (Full-time) — *January 2024 - Present*
  - **Low Code Platform (LCP) - Multimodal GraphRAG Platform**: Architected a production **multimodal RAG platform** streaming **documents, video, image, audio & text** through **Apache Kafka** ingestion pipelines into open-source LLMs (Qwen2.5-7B-Instruct, Llava-1.5-7B, nllb-200-3.3B) for speech-to-text, image/video-to-text & cross-lingual translation, fusing **Neo4j hybrid retrieval** (native vector index + LLM-constructed **knowledge-graph traversal over 50K+ documents**) via LangChain LCEL chains to serve grounded, citation-backed answers at **95%+ accuracy**
  - **High-Performance Inference Optimization**: Migrated from Ollama local inference to **vLLM on Nvidia Triton Inference Server**, implementing tensor parallelism and dynamic batching to achieve **3.2x throughput improvement** and **40% latency reduction** while supporting **concurrent processing of 50+ requests**
  - **Rust Server Migration - Performance Engineering**: Led complete server migration from Python to Rust for in-house Kepler AI platform, achieving **360% performance boost (5 FPS → 23 FPS)** and **60% memory usage reduction**. Integrated **100% of existing Python AI solutions** using PyO3 bindings and implemented dynamic configuration management with **zero-downtime deployment**
  - **Advanced Model Quantization & Cross-Platform Optimization**: Researched and implemented precision quantization strategies (FP16, BF16, INT8, INT4) for custom object detection models, achieving **4.8x faster inference**, **65% memory footprint reduction**, and **3.2x faster model load times** while maintaining **100% detection accuracy**. Optimized models for GPU/CPU/TPU/NPU using ONNX Runtime, TensorRT 8.6, and OpenVINO 2023.3
  - **Containerized GPU-Accelerated Deployment**: Developed Docker containerization for Rust-based inference pipeline with CUDA/cuDNN support, enabling **horizontal scaling across 10+ GPU instances** and reducing deployment time from **45 minutes to 8 minutes** with automated CI/CD integration
  - **Vision-Language-Action (VLA) Model Development**: Spearheading development of custom VLA model for robotics manipulation inspired by OpenVLA and LeRobot architectures. Designed end-to-end ML pipeline with custom PyTorch DataLoader supporting **multimodal data ingestion (RGB, depth, proprioception)** and implemented an efficient batching collator for **training datasets exceeding 100GB**
- **Rupeek Finance** — Bengaluru, Karnataka, India
  Data Scientist Intern (Full-time) — *June 2023 - August 2023*
  - **Multi-Algorithm Fraud Detection System**: Engineered sophisticated fraud detection pipeline using **ensemble of 3 ML algorithms** (Random Forest, XGBoost, Gradient Boosting) on **75,000+ credit reports**, achieving **96.8% precision**, **91.3% recall**, and **F1-score of 0.94** while reducing **false positives by 23%**
  - **Predictive Analytics & Default Risk Assessment**: Built multi-class classification models using Logistic Regression, Support Vector Machines, and Deep Neural Networks with attention mechanisms, achieving **AUC of 0.93** for default prediction and reducing loan approval errors by **34%**, directly impacting **11,000+ gold loan applications monthly**

Projects

- **LegalVisor - AI-Powered Legal Contract Review Platform (RAG, Hybrid Retrieval, LLM Guardrails)**: Built a production-grade RAG platform for automated Indian-law contract review across **13 contract types**, serving **Qwen2.5-7B-Instruct-AWQ on vLLM** (single 8GB GPU) behind an OpenAI-compatible API. Engineered **hybrid dense+sparse retrieval (BGE-M3, 1024-dim) with Reciprocal Rank Fusion** over Qdrant, two-stage retrieve-then-rerank via a **bge-reranker-v2-m3 cross-encoder**, layout-aware structural clause chunking, and **8 deterministic guardrails** (prompt-injection, scope, n-gram citation-grounding) enforcing hallucination-free, citation-backed JSON verdicts. Orchestrated durable ingestion with **Temporal** across a **9-service uv monorepo**. **Tech:** Python, FastAPI, vLLM, Qdrant, PostgreSQL, Temporal, Docker, Kubernetes/Helm, OpenTelemetry, Prometheus/Grafana, Next.js
- **Inferno — High-Performance LLM Inference Engine (Production-Grade, OpenAI-Compatible)**: Developed a production-grade LLM inference engine integrating advanced techniques from vLLM and SGLang for scalable, low-latency language model serving; implemented **PagedAttention and RadixTree-based prefix caching for memory-efficient KV cache management, continuous batching and speculative decoding** to boost throughput, and OpenAI-compatible REST API for seamless client integration. Engine supports multi-model inference (Llama, Mistral, Qwen, Phi) with INT8/FP8 quantization and streaming generation, enabling high-performance deployment across CPU/GPU environments. **Tech:** Python, Rust/C++ internals, REST API, quantization, high-throughput inference systems.